

MEHUL DESAI

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SUMMARY

With over 20 years of experience in science education and a solid foundation in physics, I am dedicated to crafting dynamic, engaging learning experiences. I have spearheaded the development of innovative, inquiry-based science and computer science curricula tailored for diverse age groups. My leadership extends to managing impactful professional development programs, reaching thousands of teachers and students across India, and driving the implementation of state-wide educational reforms.

As the creator of the YouTube channel "Intellectual Rigour," I share compelling stories of leading scientists with a global audience. By applying my expertise in cognitive science and instructional design, I simplify complex topics to inspire curiosity and promote lifelong learning.

PROFILE SNAPSHOT

- Designed and implemented five inquiry-based science learning programs, tailored to diverse national and international curricula including the British National Curriculum, Australian Curriculum, Cambridge International Examinations (CIE), CBSE, and CISCE. Ensured alignment with global benchmarks like the Next Generation Science Standards (NGSS) for enhanced educational outcomes.
- Created a comprehensive K–12 computer science curriculum, meticulously aligned with globally recognized standards such as the International Society for Technology in Education (ISTE) and the Computer Science Teachers Association (CSTA) to foster digital literacy and computational thinking skills.
- Played a pivotal role in educational non-profit initiatives, driving significant enhancements in classroom instruction and practices for foundational literacy and numeracy (FLN), scientific literacy, and mathematical thinking. These efforts positively impacted government schools across Gujarat, Karnataka, Telangana, and Andhra Pradesh.
- Enhanced student learning outcomes by leveraging assessment and usage data from AssessEd, providing insightful guidance to the machine learning (ML) team. This collaboration refined algorithms, ensuring a more personalized and effective learning experience.
- Spearheaded the design, development, and delivery of dynamic teacher professional development workshops, fostering expertise in inquiry-based learning, the nature of science, mathematical thinking, effective school assessments, computer science education, STEM education, and experiential learning. These workshops empowered educators to implement innovative, student-centered teaching strategies for enhanced classroom impact.
- Created a question tagging framework using Bloom's taxonomy to categorize questions, enabling the generation of insightful reports and personalized learning paths.
- Developed a comprehensive knowledge map to define optimal learning paths for students facing challenges in science and mathematics, enhancing the effectiveness of adaptive learning experiences.

EXPERIENCE

Chief Learning Officer • April 2023 - Present • Qbits Learning Pvt. Ltd., INDIA | UAE | UK | USA
Driving curriculum design, research, and development for K–12 computer science to enhance accessibility and foster mastery across diverse CS domains, empowering the next generation of innovators.

Chief Learning Officer • November 2020 - March 2023 • Yardstick Educational Initiatives, INDIA | UAE | UK | USA

Leading curriculum design, research and product development by conceptualizing learning solutions, developing frameworks, guidelines and quality assurance processes for multiple subject areas aligned to international standards for computer science, mathematics and science education.

State Teaching and Learning Lead • March 2020 - November 2020 • Central Square Foundation, New Delhi, India

Led the design and implementation of the teaching and learning modules that include working and aligning with multiple experts, partners and state government to develop a robust, scalable in-classroom instructional design and associated training elements. Worked with researchers and internal CSF research team to identify potential areas of action-research to improve the program and manage the implementation of the same.

Head of Academics - AssessEd • October 2017 - June 2019 • Digital Aristotle, Bangalore, India

Led academics by mentoring content development team, training team and operations team at AssessEd. Designed a robust architecture for content creation for learning-cum-remediation system for mathematics and science. Created entire training program and trained school teachers, Principals and other stakeholders. Stabilized operations by setting up robust sales-to-operations processes and yearly long service delivery commitments.

Project Manager - STEM Education • July 2013 - September 2017 • Fitkids Education and Training Pvt. Ltd., Bangalore, India

Conceptualized the initial development, built a team of science educators and led the implementation of DiscoverEd - an inquiry-based science learning program - in schools. Collaborated with multiple stakeholders and handled team size of close to 30 that included 15 direct reportees, handled implementation for 5 states

Sr. Subject Matter Expert - ScienceLab • October 2012 - June 2013 • Next Education India Pvt. Ltd., Hyderabad, India

Led the entire development of ScienceLab that includes conceptualization, defining program philosophy, content development, design, material sourcing, kit development and assessments. I had delivered the entire project in a short span of 8 months.

Content Lead and Subject Matter Expert (Physics) • May 2010 - September 2012 • Yardstick Educational Initiatives Pvt. Ltd., Hyderabad, India

Developed more than 200 hands-on science activities for school children. Mentored and trained the entire content development, training and program team. Trained more than 300 teachers for program implementation at schools.

Physics Teacher - Higher Secondary Division • August 2005 - April 2010 • Heaven Vidhyalaya, Vadodara, Gujarat, India

Headed the Physics Department of Higher Secondary Division. Planned the curriculum, year planner and took first hand responsibility of teaching and assessment activities. Taught Physics courses at 11th and 12th grade consisting of 200+ students every year.

EDUCATION

M.Sc. Physics (Nuclear Physics) • 2003-2005 • The Maharaja Sayajirao University of Baroda
Academic Secretary of the Physical Society at the Department of Physics, Faculty of Science

Master's Project • 2004 • Astronomy and Astrophysics Division, Physical Research Laboratory
Current research trends in the field of Astronomy

B.Sc. Physics • 2000-2003 • The Maharaja Sayajirao University of Baroda
Physics(Major), Mathematics, Statistics

TECH SKILLS

Block-based coding • ScratchJr, Scratch, MakeCode

Programming • Fortran, Python, HTML, CSS, JavaScript, Java

Data Science • NumPy, Pandas, Matplotlib, Plotly, Dart

Operating Systems •  Linux,  Mac,  Windows

Project Management • ClickUp, Todoist

PERSONAL INFORMATION

Full Name • Mehul Ratubhai Desai

Date of Birth • 9th February, 1982

Marital Status • Married

Language Proficiency • English, Gujarati, Hindi

PROJECTS

Qubits CS • September 2021 - Present • Qbits Education Pvt. Ltd. (a sister concern of Yardstick)

Leading the curriculum design for futuristic computer science curriculum for k-12 aligned to international standards such as International Society for Technology in Education(ISTE) and Computer Science Teachers Association(CSTA). More information: <https://qubitscs.com>

BeeCurious • November 2020 - March 2023 • Yardstick Educational Initiatives Pvt. Ltd.

Leading the curriculum design for inquiry-based science modules developed for international standards such as Next Generation Science Standards (NGSS), British National Curriculum, Cambridge International Examinations(CIE) and Australian Curriculum. More information: <https://www.beecuriousedu.com>

Roost at Home • May 2020 • Central Square Foundation

Led the initiative "Roost at home", aimed to provide psychosocial support and socio emotional learning for all English medium schools across the state of Gujarat amid COVID-19 crisis. Gujarat council of Educational Research and Training (GCERT) disseminated the learning packs through their cascaded WhatsApp groups. Collaborated with multiple NGO partners to make the initiative successful.

AssessEd • May 2016 - July 2019 • Digital Aristotle Pvt. Ltd.

In the reverse chronological order (a) Developed a content framework for revision and remediation system based on Revised Bloom's Taxonomy. (b) Mentored a team of 12 interns to create the base for an Intelligent Tutoring System (ITS) for science and math subjects. (c) Developed the entire training program for implementing AssessEd smoothly in Schools. (d) Analyzed the data of students and their assessments, generating insightful reports for all stakeholders and guide the pedagogy (e) Initiated a new structure for assessment bank development.

Eureka 5.0 • August 2017 - April 2019 • Thinklabs Technosolutions Pvt. Ltd. (Fitkids)

Played a vital role during the due diligence stage of acquisition Thinklabs by Fitkids. During the above mentioned period, laid out the content guidelines and structure for the revival of Eureka 4.0 program to Eureka 5.0. Reviewed and created content for the project. Also mentored the entire Eureka team of trainers. Solely developed the entire supply chain management system, by creating and connecting google spreadsheets that served as an automated system for handling the whole supply chain. This supply chain delivered over six lacs kits to 80+ schools across India. Eureka - Active Science Learning - integrates 5E learning methodology.

DiscoverEd • July 2013 - April 2017 • Fitkids Education & Training Pvt. Ltd.

DiscoverEd is a program aligned to Next Generation Science Standards (NGSS). The program aims to develop science process skills and knowledge by engaging students in science practices. Students use 'Claim-Evidence-Reasoning' as a framework to explain their finding from the investigations. The programme aims to focus on 'doing science' as opposed to 'learning science' passively. I was instrumental in implementing the program by designing the entire operations of DiscoverEd Program.

Project Ramanujan • March 2015 - March 2016 • Fitkids Education & Training Pvt. Ltd.

This research project aimed to understand the challenges faced by Math teachers; and identifying the lacuna lying in math education. It endeavoured to find successful teaching pedagogy that can effectively develop students' mathematical thinking skills. Led this project by planning the pedagogy, developed the lesson plans, conducted and evaluated diagnostic tests and implemented the lessons myself by teaching at The King's School, Goa. Also administered a math perception survey before and after the entire intervention; analyzed and presented the data to the stakeholders.

Project Chandrasekhar • July 2013 - January 2014 • Fitkids Education & Training Pvt. Ltd.

*As a part of this project, I had evaluated various robotics kits such as **LEGO Mindstorm**, **VEX**, **Fischertechnik** in a hands-on manner to review its appropriateness for the integrated STEM curriculum. Developed the basic framework of the STEM program by performing the initial examination of the prototype curriculum inhouse. The effort in this project led to the final development of DiscoverEd program.*

Project Feynman • July 2013 - January 2014 • Fitkids Education & Training Pvt. Ltd.

*As a part of this project, I had collaborated with pedagogy team from world-leading organizations in science education such as **LaB-aiDS**, **STEMscopes** (RICE University), **Carolina**. This project endeavoured to draw their learning to create an integrated STEM Education program for Indian Schools. The efforts from this project contributed critical inputs to the final development of DiscoverEd program.*

ScienceLab • October 2012 - June 2013 • Next Education India Pvt. Ltd.

Science Lab is in 70+ schools in India. The program is aligned to NCF-2005 and aims to develop curiosity by experiential science activities. The modular kits help students to perform ten hands-on activities in each grade. In a short span of 8 months, I had created the entire science kit with content starting from the conceptualization till the go-to-market stage. More information about ScienceLab is available at the official website: <http://nextsciencelab.in/>

Yardstick Experiential Learning Program (YELP) • May 2010 - September 2012 • Yardstick Educational Initiatives Pvt. Ltd.

One of the 1st hands-on science program, in India, based on David Kolb's Experiential Learning Cycle. The program aims to provide enriching experience to students by engaging them into hands-on inquiry. I had developed more than 200 activities for school kids for Physics concepts and reviewed the content of more than 200 activities for Chemistry and Biology. Developed the teacher training program and implemented it across 30+ schools. Taught hands-on activities in 3 schools at Hyderabad. More information about YELP is available at the official website: <https://www.yardstickedu.com>

School Excellence Program • January 2011 - April 2012 • Yardstick Educational Initiatives Pvt. Ltd.

The project was aimed to improve quality of science education in government schools. We successfully implemented the project in Hyderabad region benefiting around 2000 students coming from families with low-income. I led the development, design and training team to build a science teaching program using demonstration activities in the classroom and train school teacher to implement it.

A short term project on General Astronomy • November, 2004 • Astronomy & Astrophysics Division, Physical Research Laboratory (PRL), Ahmedabad, India

As a part of this short-term project, I had interacted with the faculty of A&A division. Our task included solving a few problems in general astronomy along with understanding the research directions pursued by the faculty at the institute.

TRAININGS CONDUCTED

Candour5 Education Academy, Bangalore • June 11 2022

VAHE Global Academy, Bangalore • June 2 and 6 2022

Howard Public School, Himayatnagar, Hyderabad • June 11 2019

Sanghamitra School, Nizampet Rd, Hyderabad • June 10 2019

VAV International School, Udumalaipettai, TN • 2019 May 27

Tapovan School, Bangalore • 2019 May 20, 21

Saraswati Senior Secondary School, Narnaul, Haryana • 2019 May 4

Gurukulam The School • 2019 May 3

Delhi Public School, Rudrapur • 2019 April 3

Aurum The Global School, Haldwani, Uttarakhand • 2019 April 2

Bhartiyam International School, Rudrapur-Kichha road, Uttarakhand • April 1 2019

Insight Academy, Bangalore • March 6 2019

Stepping Stones Senior Secondary School, Chandigarh • February 21 2019

Vivekanand School, Anand Vihar, New Delhi • February 20 2019

Venkateshwar Global School, Sector 13, Rohini, New Delhi •

Genesis Global School, Noida • February 16 2019

Shri Venkateshwar International School, Sector 18, new Delhi • February 13 2019

Tapovan School, Koppa Gate, Bangalore, Karnataka • January 28 2019

Sri Sankara Senior Secondary School, Arunachalapuram, Chennai, Tamil Nadu • January 12 2019

Shri R. M. Jain Vidhyashram, Kakkalur, Chennai, TN • January 10 2019

St. Joseph's School, Habsiguda, Hyderabad • December 15 2010

Kotwal School, Maredpally, Hyderabad • December 18 2010

Kotwal School, Bolaram, Hyderabad • December 20 2010
 Mother's Integral, Hyderabad • December 21 2010
 Auden Public School, Giri Nagar • January 2011
 Johnson Grammar School, Hyderabad • April 17-18 2011
 Indus World School, Indore • May 17-22 2011
 30 teachers Carmel High School, Bangalore • May 25 & 27 2011
 Little Flower School, Bangalore • May 28, 30 & 31 2011
 DAV Public School, Bangalore • June 1-2 2011
 Pragati Vidyaniketan School, Amberpet • June 10 2011
 Sharada High School, Iasnapur • June 11 2011
 St. Joseph's School, Malakpet • June 11 2011
 St. Joseph's School, Habsiguda • June 14 2011
 Delhi Public School, North Bangalore • June 29-30 2011
 Kotwal School, Bolaram, Hyderabad • July 1 2011
 Genesis International School, Miyapur • June 18 & 25 2011
 Martin Luther Public School • May 2011
 Johnson Grammar School, LB Nagar • July 9 2011
 Indus World School, Ahmednagar • July 30-31 2011
 St. Andrews School, Hyderabad • August 23 2011
 Indus World School, Indore • November 18-19 2011
 St. Andrews School, Hyderabad • December 10 2011
 St. Andrews School, Hyderabad • January 12-13 2012
 Gitanjali Devshala, Hyderabad • November 28-29 2011
 Johnson Grammar School, Nacharam • December 14-16 2011
 DPS North • May 28-29 2012
 DAV Public School • May 30-31 2012
 Carmel High School • June 1-2 2012
 Sri Kumarans ICSE & CBSE • June 7-8 2012
 Modern School, New Delhi • April 13-14 2012
 St. Andrews, Bowenpally • April 18 2012
 Little Flower School, Bangalore • May 26 & June 9 2012
 Johnson Grammar School, Habsiguda • April 20-21 2012
 St. Joseph's School, Habsiguda • April 23 2012
 St. Joseph's School, Malakpet • April 25-26 2012
 Kamala Niketan, Trichy • May 28-29 2012
 DAV Public School, Bangalore • May 30-31 2012
 Carmel School, Bangalore • June 1-2 2012
 Sri Kumaran's Children's Home, Bangalore • June 7-8 2012
 Little Flower Public School, Bangalore • June 9 2012